

A High Speed CMOS Amplifier with Active Feedforward Compensation for Contactless Card Reader

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Abstract

A high speed CMOS two stage amplifier with active feedforward compensation is designed for the 13.56MHz type B contactless card reader whose data rate is 847 kHz. A rail-to-rail input folded cascode amplifier is used as the first stage, a differential push-pull amplifier as the second stage, and a push-pull amplifier as the feedforward compensation stage. The active feedforward compensation shows better frequency character than the traditional compensation schemes, and the push-pull structure promotes the performance of amplifier. The amplifier was simulated in HHNEC 0.35 μ m CMOS digital process with single 3.3V power supply. The simulation results showed dc gain of around 93dB and phase margin of around 50 degree with high driven ability of a 20 PF capacitor. The gain bandwidth (GBW) is above 1GHz at the signal frequency 847 kHz, while its unit gain frequency is only 340MHz. This character is very useful to restrict the higher frequency carrier wave signal. It has been applied in the preamplifier, active RC filter and SC filter in our application.

1. Introduction

High performance amplifiers with both high gain and high speed are increasingly demanded in modern communication application. In order to satisfy the monolithic requirements, analog IC designers are confronted with great challenges of designing high gain-bandwidth product (GBW) amplifiers in traditional digital CMOS process. Various attempts have been developed to satisfy the requirements. In order to get a high gain, cascoding, telescopic and gain-boosting techniques are used in [1] [2] [3] [4] [5]. Cascoding is a well-known technique that increases the open loop gain of the operational amplifier without degrading the high-frequency performance. Telescopic structure provides medium gain but shows higher speed, low noise and low power dissipation characters. Gain-boosting techniques that utilize a full opamp can increase the dc gain of the operational amplifier. On the other hand, since traditional Miller compensation method often shows bad characters of high frequency, advanced compensation schemes have been proposed to enhance the bandwidth of the amplifiers and ensure amplifiers' stability. Damping-factor control frequency compensation (DFCFC) [6], [7] and positive feedback com-

pensation (PFC) [8] enhancing the bandwidth by removing the inner Miller capacitor loaded at the amplifier output. Active feedback frequency compensation (AFFC) [9] uses an active capacitive feedback network to create a high speed feedback path for high frequency signal propagation, and thus improve the bandwidth. Dual-path amplifier topology with dual-loop parallel compensation (DLPC) technique [10] and multipath amplifier [11] were also proposed to promote the bandwidth. Extraordinarily, feedforward compensation scheme with no Miller capacitors [12] has shown better trade-off between frequency and gain character with simple compensation circuits.

In this paper, a more effective circuit implementation is proposed based upon the basic idea of feedforward compensation. We use a rail-to-rail input folded cascode amplifier as the first stage, a push-pull structure as the feedforward compensation stage. The active feedforward compensation shows better frequency character for the high speed application than the traditional compensation schemes, and the push-pull structure promotes the performance of the amplifier.

2. Two Stage High Speed Amplifier with Active Feedforward Compensation

2.1 The circuit principle

High speed communication applications propose requirements for amplifiers with high gain at high signal frequency. But it is difficult to design such an amplifier with good stability, since stability contradicts with GBW. One of the key concerns in the designs is phase margin compensation. Despite compensation capacitor can obviously enhance the stability of amplifiers, it degrades amplifier's high frequency performance seriously, which is the main obstacle to high speed communication application. The new feedforward compensation idea with no miller capacitors [12] is very helpful to solve this problem.

As shown in figure 1, we use the positive phase shift of left-half-plane (LHP) zeros caused by the feedforward path of A3 to cancel the negative phase shift of poles caused by A1 and A2 [12]. The dominant pole is not pushed to lower frequency, resulting in a higher gain-bandwidth product with a fast response. The overall transform function of the amplifier is shown as

$$H(s) \approx [A_1 A_2 + A_3(1+s/\omega_z)] / [(1+s/\omega_{p1})(1+s/\omega_{p2})] \quad (1)$$

In the experiments, the idea of the active feedforward was promoted by a push-pull structure to achieve a good stability. The rail-to-rail folded cascode input stage also contributes to better frequency character and stability.

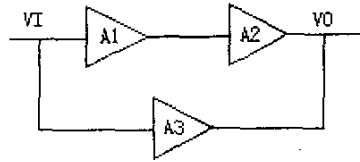


Figure 1. The block diagram of the amplifier

2.2 The implementation of the amplifier

The first gain stage is shown in figure 2. The rail-to-rail input stage composed of M1PA, M1PB, M1NA, M1NB, is designed to obtain a constant G_m independent of common mode input voltage. M2A, M2B, M3A, M3B, M4A, M4B, M5A, M5B compose of the folded cascode structure. MCM1A, MCM2A, MCM1B, MCM2B form a common-mode feedback (CMFB) circuit to increase the stability of the amplifier. Transistors MB1N, MB1P generate the bias current to drive the input stage. The bias voltage is not generated from external circuit, but from the common-mode output end VO1A. The transform function of A1 is described as

$$A_{v1} = (g_{m1nb} + g_{m1pb}) [g_{m4b} r_{ds4b} r_{ds5b} || g_{m3b} r_{ds3b} r_{ds2b}] \quad (2)$$

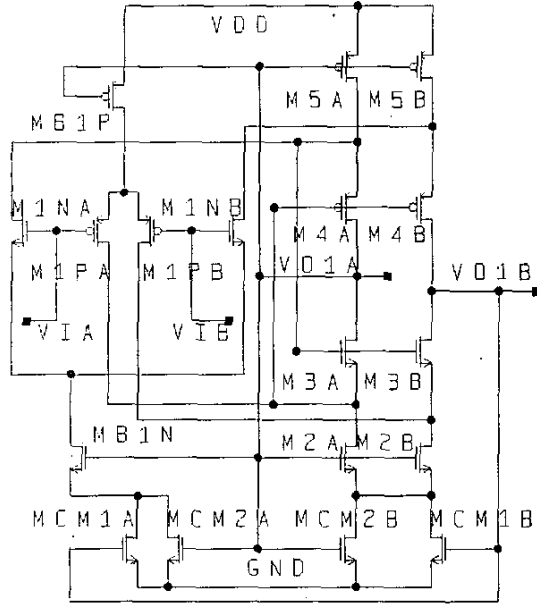


Figure 2. The first stage of the amplifier

The second stage is a push-pull two-ended input sing-ended output amplifier as shown in figure 3. Transistors M6A, M6B, M7A, M7B form the push-pull gain stage, and current mirror composed of M8A, M8B

ensures the positive gain of the second stage. The push-pull structure can increase the gain and robust characters with lower (W/L) ratios of transistors.

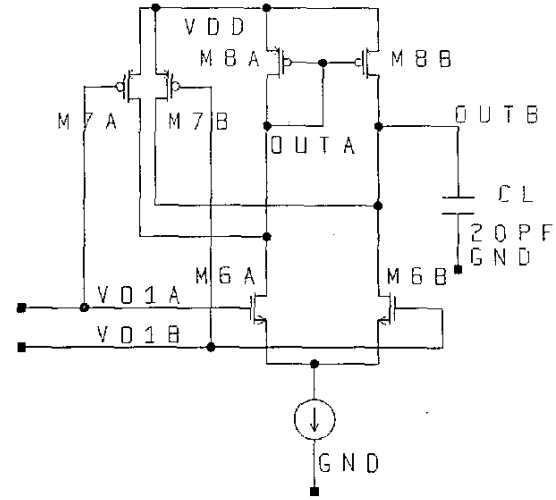


Figure 3. The push-pull second gain stage

The compensation stage A3 is similar to A2, as shown in figure 4. Transistors M9A, M9B, M10A and M10B form the push-pull active feedforward compensation path. The push-pull structure can provide more reliable and efficient compensation in contrast to the compensation form in [12]. The gain of A2 can be described as

$$A_{v2} = (g_{m6b} + g_{m7b}) [r_{ds6b} || r_{ds7b} || r_{ds8b} || r_{ds9b} || r_{ds10b}] \quad (3)$$

And the transform function of A3 is

$$A_{v3} = -(g_{m9b} + g_{m10b}) [r_{ds6b} || r_{ds7b} || r_{ds8b} || r_{ds9b} || r_{ds10b}] \quad (4)$$

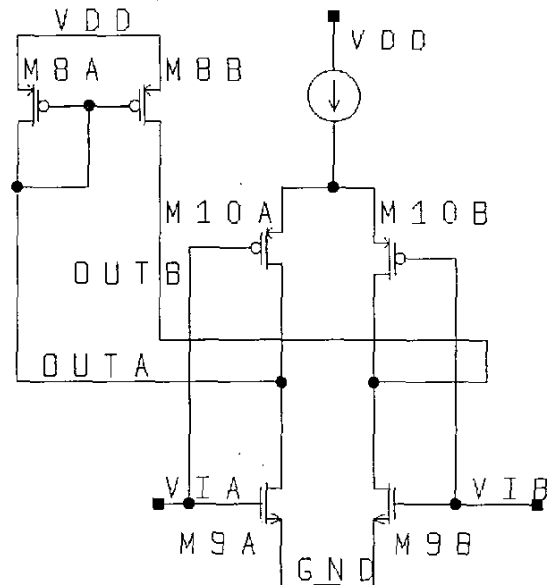


Figure 4. The push-pull feedforward compensation stage

2.3. Some design tips

There are some design tips to be noticed. Firstly, this amplifier use VIA as the common mode end and VIB as the signal end, the common mode voltage can be regulated according to the requirements of the application. It was set at 1.45V in the experiments. Secondly, the bias current of the second stage should be balanced between the gain and drive ability. To get a higher gain, it should be smaller. The reason relies on that the output resistance increases in proportion to a decrease in bias current, whereas the transconductance increases as the square root of the increase in bias current. But the needed driven ability requires a higher current. On the other hand, the bias current of the compensation stage should be higher so as to increase the transconductance of M9A, M9B, M10A and M10B, thus increases the phase margin of the amplifier. Since the above four transistors' (W/L) ratios are a little larger, we realized the differential pair with "common-centroid" and interdigitated layout. Thirdly, the CMFB transistors MCM1A, MCM2A, MCM1B and MCM2B are helpful to the amplifier's stability. The amplifier without the CMFB transistors was also simulated, and the first stage's output VO1B was found unstable, thus degraded the stability of the whole amplifier. Besides, it is helpful to settle (W/L) ratios of M10A and M10B small, and we set (26/1) in the simulations.

3. The Simulation Results

The active feedforward compensation two stages amplifier has been simulated in 0.35 μ m HHNEC digital CMOS process with single 3.3 V supply voltage. The amplifier is loaded with a 20pF capacitor in order to satisfy the requirements of the preamplifier, active RC filter and sc filter for the wireless card reader's application.

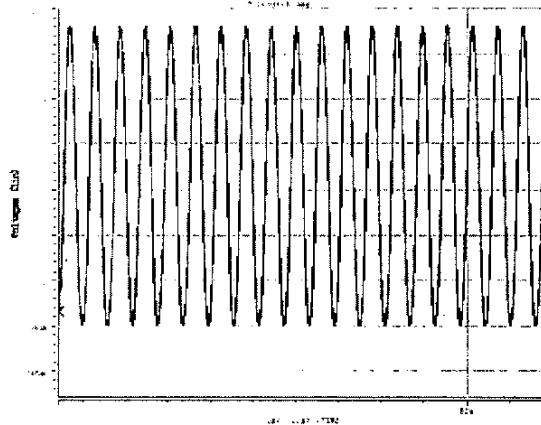


Figure 5. The transient analysis wave of OUTB
In the experiments, a lot of simulations were made

towards the variable power supply voltage ranges from 3.0V to 3.6V, the variable temperature ranges from -20°C to 80 °C, and different combinations of FF, FS, SF, SS, TT MOSFET models. At last, we got a group of trade-off parameters among these variable factors. The amplifier has about a DC gain of 93dB and AC gain of 62dB at the signal frequency of 847 kHz in the typical conditions. The result is important for the short distance contactless card reader application, since the signal amplitude is very weak but the data rate is high— 847 kHz.

The transient analysis wave of OUTB is shown in figure 5 when the amplitude of the small sine waves input signal is 0.5mv and the frequency is 847 kHz. We get a high gain bandwidth of above 1GHz at the signal frequency of 847 kHz which suits to our application. The AC simulation result in figure 6 shows that the amplifier has a dc gain of more than 93dB and a unit gain frequency of 340MHz. The relatively low unit gain frequency is also helpful for the short wireless communication since it acts to restrict the carrier wave signal. The phase response of open-loop in figure 7 shows that the amplifier has a phase margin about 50°. So it is stable for the feedback application.

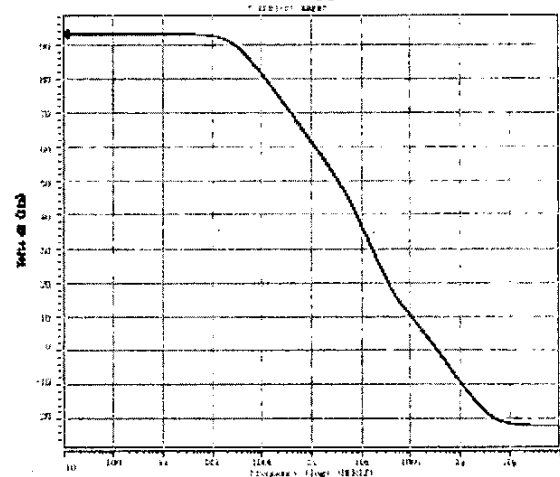


Figure 6. The open-loop ac amplitude response

4. Summary

This paper presents a new design of a high speed CMOS amplifier for the short distance contactless card reader. It uses a folded cascode gain stage and active feedforward compensation by a push-pull structure to get a high gain at the signal frequency of 847 kHz and a good stability for the feedback application. Simulation results show that it has a GBW of above 1GHz at this frequency with a load capacitor of 20 PF. The phase margin is about 50°. It has been applied in the preamplifier, active RC filter and SC filter for the contactless card reader application.

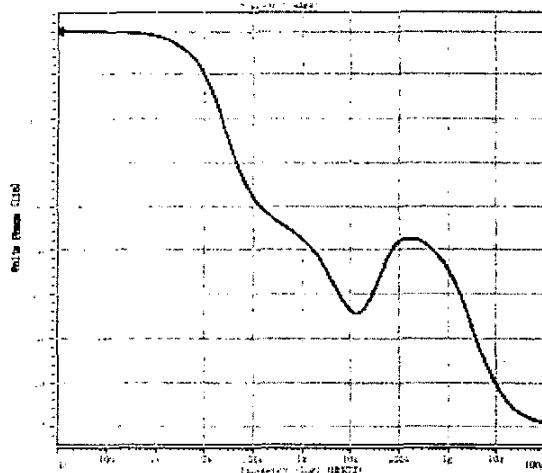


Figure 7. The open-loop ac phase response

References

- [1] A. Younis and M. Hassoun, "A High Speed Fully Differential CMOS Opamp," Pro. 43rd IEEE Midwest Symp on circuits and systems, Lansing MI, pp.780-783, (2000).
- [2] K. Bult and G. J. G. M. Geelen, "A fast-settling CMOS op amp for SC circuits with 90-dB DC gain," IEEE Journal of Solid-State Circuits, pp.1379-1384, (1990).
- [3] R. Sun and L. Peng, "A gain-enhanced two-stage fully-differential CMOS op amp with high unity-gain bandwidth," IEEE Journal of Solid-State Circuits, pp.428-431, (2002).
- [4] M. M. Amourah and R. L. Geiger, "Gain and bandwidth boosting techniques for high-speed operational amplifiers," IEEE Journal of Solid-State Circuits, pp.232-235, (2001).
- [5] B. G. Song, O. J. Kwon, I.K. Chang, H. J. SONG and K.D. Kwack, "A 1.8 V self-biased complementary folded cascode amplifier," IEEE Journal of Solid-State Circuits, pp.63-65, (1999).
- [6] K. N. Leung, P. K. T. Mok, W. H. Ki, and J. K. O. Sin, "Three-stage large capacitive load amplifier with damping-factor-control frequency compensation," IEEE J. Solid-State Circuits, vol. 35, pp. 221-230, (2000).
- [7] K. N. Leung and P. K. T. Mok, "Analysis of multistage amplifier-frequency compensation," IEEE Trans. Circuits Syst. I, vol. 48, pp.1041-1056, (2001).
- [8] J. Ramos and M. Steyaert, "Three-stage amplifier with positive feedback compensation scheme," in Proc. IEEE Custom Integrated Circuits Conf., Orlando, FL, pp. 333-336, (2002).
- [9] H. Lee and P. K. T. Mok, "Active-feedback frequency-compensation technique for low-power multistage amplifiers," IEEE J. Solid-State Circuits, vol.

38, pp. 511-520, (2003).

[10] Hoi Lee Ka Nang Leung and Philip K. T. Mok "A dual-path bandwidth extension amplifier topology with dual-loop parallel compensation," IEEE Journal of Solid-State Circuits, pp.1739-1744, (2003).

[11] M. Schlarmann, E. Lee, and R. Geiger, "A new multipath amplifier design technique for enhancing gain and bandwidth," in *International Circuits Symposium*, vol. II, pp. 610-615, (1999).

[12] B. K. Thandri and J. Silva-Martinez, "A Robust Feedforward Compensation Scheme for Multistage Operational Transconductance Amplifiers With No Miller Capacitors," IEEE Journal of Solid-State Circuits, Vol. 38, pp.237-243, (2003).